A New Object-Oriented SAR Interferometry Framework for Monitoring Urban Deformation

Chisheng Wang, Chuanhua Zhu, Xiang–Sheng Wang, Wei Tu, *Member, IEEE*, and Qingquan Li

***Abstract*— Precise deformation surveying is essential for ensuring safety and sustainable development in densely populated cities. Synthetic aperture radar interferometry (InSAR) is a technique well suited for large-scale deformation measurement. It is widely applied in monitoring geological disasters such as earthquakes, landslides, and volcanoes. However, it faces challenges when used to monitor urban infrastructure, which requires higher precision and more details. To overcome these challenges, we introduce an innovative, object-oriented multitemporal InSAR (Obj-InSAR) framework. Compared with state-of-the-art InSAR methods, Obj-InSAR shifts the basic processing unit from pixel to object. This allows the use of additional information provided by the object/segment to enable detailed and precise results, as required in urban scenarios. In this framework, a novel object-based plug-in network is designed to establish an impressively dense network for parameters solving. Additionally, a new object-based adjustment procedure through 3-D point cloud alignment is proposed to accurately determine the location of InSAR scatterers and guide network construction. As a result, our approach produces a more detailed, accurate and precisely geolocated 4-D infrastructure point cloud than conventional pixel-oriented frameworks, enabling effective monitoring of urban deformation. A case study confirms the usefulness and adaptability of this approach in urban SAR interferometry. These advancements will contribute to the development of time-series InSAR technology in the new era, enhance the capability of monitoring complex urban deformation in high-density cities, and serve urban safety operations and sustainable economic development.**

***Index Terms*— Deformation monitoring, object-oriented, urban infrastructure, synthetic aperture radar interferometry (InSAR).**

I. Introduction

CITIES and metropolitan areas contribute significantly to economic growth and technology innovations. Worldwide urbanization across the globe results in a large number of people migrating into cities. This trend is expected to continue, and it is predicted that by 2030, 60% of global population will reside in cities [1]. Urban infrastructure, including buildings, roads, and bridges, plays a crucial role in managing the strain caused by rapid population growth [2]. Thus, ensuring the safety of infrastructure is essential for sustainable cities and communities, aligning with Sustainable Development Goals established by the United Nations [3]. However, various man-made and natural hazards can impact the health of urban infrastructure, potentially leading to catastrophic failures. It is therefore critical to regularly monitor infrastructure deformation over a wide area with high resolution.

Multitemporal synthetic aperture radar interferometry (MT-InSAR) is a commonly used technique for measuring ground deformation [4]. With the increasing spatial resolution and data availability of SAR images, many studies have been employing MT-InSAR for infrastructure monitoring [5], [6]. Compared to traditional field surveying, MT-InSAR offers advantages such as large-scale monitoring capability, dense point sampling, and noncontact measurement [7], [8]. There are essentially two stages in the InSAR algorithm development. The first stage involves image-based processing, where the basic data unit is the image. Differential interferometry, unwrapping, filtering, and parameter estimation are all performed on the images. This processing method remains popular due to its simplicity and fast computation. Conventional D-InSAR [9], most SBAS [10], [11], [12], and stacking algorithms [13] fall into this category. However, the image-based processing method has some drawbacks. It relies on grid operation and cannot distinguish between low-quality and high-quality pixels with fine resolution. Therefore, the spatial resolution and precision of the results are limited. The second stage involves pixel-based processing. Here, instead of processing the entire image, only pixels with high quality are selected for further processing [14]. Since we only focus on high-quality pixels, multilook filtering is not required, and the maximum resolution can be retained. Moreover, contamination from neighboring low-quality pixels is minimized. Pixel-based algorithms are widely used in the InSAR community, with most state-of-the-art algorithms falling into this category, including IPTA [15], STAMPS [16], STUN [17], PSPs [18], Tomo-GENESIS [19], and SqueeSAR [20], [21].

Measuring urban infrastructure deformation with MT-InSAR is considerably more challenging because of the complex urban environment and specific engineering standards. On the one hand, urban environments are mixed

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with buildings, railways, roads, trees, waters, etc. Many types of scatterers exist, including distributed scatterers, persistent scatterers, overlaying scatterers, shadows, and so forth. In addition, common assumptions based on homogeneous characteristics, like similar phase patterns between neighboring pixels, are invalidated. On the other hand, infrastructure safety assessments require fine interpretation of deformation, necessitating higher point density and more accurate positioning and deformation accuracy from MT-InSAR measurements. Thus, measuring urban infrastructure deformation using MT-InSAR remains a significant challenge.

After continuous development of MT-InSAR techniques over the last few decades, the information contained in SAR pixels has been fully exploited. The precision of results produced by existing algorithms has nearly reached the upper limit of the available SAR images. To further improve these results, we must incorporate more information beyond what pixels can provide. Object-oriented image analysis is a significant advancement in the field of remote sensing image processing. It offers a critical bridge between spatial concepts in geographic information science and radiometric observations in remote sensing images. Compared to pixel-by-pixel processing, an object-oriented approach can leverage additional information provided by the object/segment, in both spectral (mean values, median values, variance, etc.) and spatial dimensions (distances, neighborhood, topology, etc.) [22], [23]. It has been claimed that segmentation in the object-oriented approach can add crucial information for remote sensing processing. Therefore, it is expected that the additional information embedded in the object can significantly improve the existing MT-InSAR performance, warranting fine and accurate measurements, as demanded by urban infrastructure deformation monitoring. Uses for object-oriented image analysis in remote sensing are numerous, but to our knowledge, little work has been done in SAR interferometry.

Our aim is to fill this gap by introducing an object-oriented MT-InSAR (Obj-InSAR) framework to extract fine and precise deformation information for urban infrastructure monitoring. More precisely, we first segment the 3-D urban structure with deep learning foundation model. We link the SAR pixels to the 3-D real world with a novel 3-D look-up table, allowing the segmentation of the 3-D model in geographic coordinates to be directly applied to SAR pixels. Subsequently, we construct an object-by-object network to estimate the required height and deformation parameters. Next, the estimated geometry of the object is registered to an external 3-D model for outlier detection and fine processing. Finally, a refined time-series analysis step is conducted once more on the segmented objects to retrieve the nonlinear deformation component. This framework enables the integration of external spatial datasets (e.g., 3-D models) for ensemble SAR interferometry processing. By properly using the additional information provided by segmented objects, it can generate a much denser and more precise 4-D infrastructure point cloud than the pixel-based approach. Also, the final monitoring result is directly linked to the object, making it much easier to manage and integrate into existing geographical information systems. The novelty and distinction from conventional pixel-based processing chains lie in three key aspects: 1) 3-D point cloud data can be integrated to correct improper network construction; 2) it offers a flexible processing strategy by focusing on objects rather than the entire area, which can drastically reduce computing time and memory consumption; and 3) the object-oriented approach allows for the construction of an extremely dense network, increasing the density of InSAR measurements.

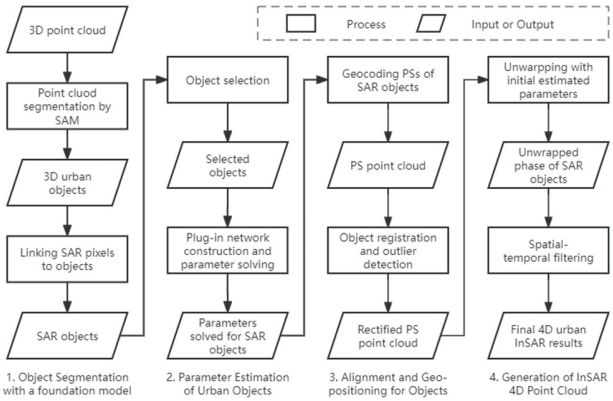


Fig. 1. Flowchart of the object-oriented SAR interferometry method.

II. Methods

In the Obj-InSAR processing scheme, we take the urban 3-D point cloud and satellite SAR images as input data and output the deformation monitoring results of the urban infrastructures in terms of a 4-D point cloud (3-D in space and 1-D in deformation). The process is divided into four main steps (see Fig. 1).

A. Object Segmentation With a Foundation Model

The first challenge of the Obj-InSAR is to segment and isolate facility objects from SAR images. Due to the SAR imaging principle, the adjacency relationship of pixels on the radar coordinate system does not align with the real world. There are phenomena unique to SAR imaging, such as overlap, shrinkage, and shadow that is not found in optical imaging. Thus, direct object segmentation methods based on SAR images are exceedingly challenging. However, extracting objects using external 3-D point clouds in the geographic coordinate system is more intuitive and accurate. Here, we proposed to identify objects first on the geographic coordinate system, and then associate the identified objects to the SAR pixels.

Recent advancements in deep learning have demonstrated the potential of large-scale pretrained models that can learn from vast amounts of data and perform various tasks with minimal supervision. Therefore, we choose to use a vision foundation model, the segment anything model (SAM), to aid object segmentation [24]. SAM is a new AI model from Meta AI that can "cut out" any object, in any image. In this study, a colorized orthographic map generated by the 3-D point cloud using a color ramp shader was used as the external auxiliary data to generate objects at all scales. The map is then fed into the SAM. To achieve practicality in urban scenarios, objects at various scales can be obtained after flexible merging or grouping according to different InSAR observation purposes.

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To associate the spatial point in the geographic coordinate system to SAR pixels in the SAR Doppler-range coordinate, we propose using a 3-D lookup table rather than the commonly used traditional 2-D lookup table. The fundamental idea is that the relationship is established between 3-D locations in geographic coordinates and 2-D pixel locations in SAR image. This extra design is crucial in complex urban scenarios because many facade details will be omitted in a 2-D DSM, causing difficulties in geolocating the SAR pixels when using a 2-D lookup table. Specifically, we establish a mesh model based on an external city 3-D point cloud and perform 3-D uniform sampling on the mesh model. For each 3-D point, we calculate the corresponding pixel on the radar image based on radar imaging geometry. The relational information, connecting 3-D geographic locations and 2-D SAR pixels, is stored in the form of a lookup table. The lookup table is used to find the corresponding radar pixels for each object. To mitigate potential errors stemming from uncertainty in SAR imaging geometry, we can run a coarse height estimation of a small set of PS and register the retrieved 3-D positions of these PSs to the LiDAR point cloud. The registration information can be used to refine the lookup table. Detailed information about the 3-D lookup table can be found in [25].

B. Parameter Estimation of Urban Objects

Next, we estimate parameters on PSs within objects. The entire process is operated at full scale without multilooking. To eliminate nonrelevant spatially correlated signals such as atmospheric noise, a spatial network needs to be constructed and solved. We use a plug-in design for network construction. A global sparse network is constructed first, and the object network is treated as an independent module that can be plugged into the global network as needed. To construct the dense network, we employ an amplitude dispersion and average intensity constraint to identify candidate PSs. We then assemble a hybrid network composed of a multilayer Delaunay network (global network) and a finite-distance fully connected network (object network) (see Fig. 2). The multilayer Delaunay network results from combining multiple Delaunay networks, each constructed using a distinct set of randomly selected points. Such a design is implemented to avoid a single Delaunay network being too sparse, which cannot provide solid constraints for global PSs. The finite-distance fully connected network connects all potential edges within a defined distance threshold. This design allows for all possible connections between PSs in selected objects. Meanwhile, incorporating a distance parameter enables the adjustment of network edge number by selecting only those edges within the permitted range.

After the network construction, parameter estimation for PSs is transferred to arcs. This estimation is implemented through periodogram optimization [14]

$$\hat{\boldsymbol{m}} = \arg\max\left\{\left|\frac{1}{n}\sum_{k=1}^{n} \boldsymbol{g}_k \boldsymbol{\Psi}_k^H(\boldsymbol{m})\right|\right\}. \tag{1}$$

Here, n is the number of interferograms, g is the time-series SAR complex value observation of the arc, $\mathbf{m}$ is the parameter vector to be estimated for the arc, for example, height, thermal coefficient, and linear velocity, and $\boldsymbol{\Psi}(.)$ is the phase model

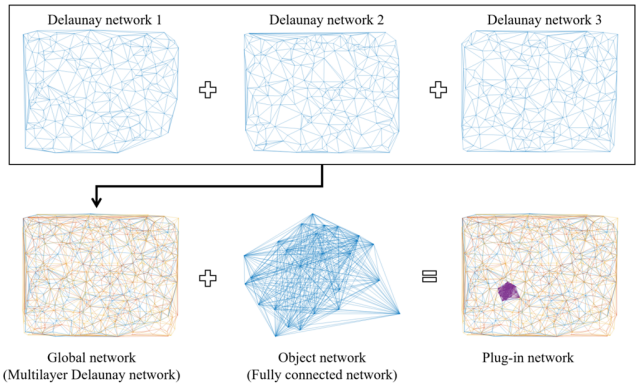


Fig. 2. Schematic of the hybrid network design.

given the parameters. Since the periodogram optimization is conducted arc by arc, we employ parallel computing to accelerate the optimization process.

To perform periodogram optimization, we define a range of values for parameters such as height, velocity, and thermal expansion coefficient, and then discretize within this range. The discretized parameters are used in (1) for calculation, and the solution corresponding to the highest periodogram value is considered the initial solution. These initial estimates are used for temporal unwrapping of the time series phase, which involves unwrapping the residuals of the observed phase after subtracting the model values corresponding to these parameters, and then adding the model values back. The phase, after temporal unwrapping, along with the parameters, forms a system of linear equations that can be solved using least squares estimation or robust least squares estimation. After solving each arc, its temporal coherence is calculated, which is used to assess the quality of the solution for that arc. If the temporal coherence is low (e.g., <0.7), the arc is removed from the network and no longer included in subsequent calculations.

The parameter on the high coherent arcs can be integrated into PSs through network adjustment. The function model between arc parameter $\Delta\boldsymbol{x}$ and PS parameter $\boldsymbol{x}$ can be described by a linear operation

$$\Delta\boldsymbol{x} = \boldsymbol{G}\boldsymbol{x}, \quad \text{where}$$
$$\boldsymbol{G} = \begin{bmatrix} 1 & \cdots & -1 & \cdots & 0 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \cdots & 1 & -1 & \cdots & 0 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & 0 & 0 & \cdots & -1 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \end{bmatrix}. \tag{2}$$

Here, $\boldsymbol{G}$ is the design matrix that connects arcs and PSs. The row size equals the number of arcs, and the column size equals the number of PSs. Network adjustment can be achieved through a least squares estimator or a robust estimator to solve parameters in PSs. However, the number of edges can sometimes be very large, making the network adjustment process impractical. To address this, a sparse matrix is used to store the design matrix. LU factorization is then employed to decompose the matrix into simpler parts for least squares

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solving [26]. Such network adjustments can be run iteratively. Arcs with high residuals can be deleted in each iteration, and the process will continue until the residuals reach a low level.

C. Fine Alignment and Geopositioning for Objects

After the elevations of the scatterers within the urban objects have been determined, they can be converted to the geographic coordinate system and aligned with the external 3-D point cloud. Based on the given SAR imaging geometry, we establish a transformation function from the scattering points in the radar coordinate system to a local Cartesian coordinate system. Assuming the pixel coordinates of a scatterer in range-Doppler coordinate is (r, a) and its estimated height is h, the coordinates in the local Cartesian coordinate system can be given by

$$\begin{aligned} x &= r \times s_r \\ y &= a \times s_a + h/\tan(\theta), \quad z = h \end{aligned} \tag{3}$$

where s_r and s_a are the spacing in range and azimuth, and θ is the incidence angle.

We calculate the point cloud position corresponding to different elevation values and align it with the external point cloud to search for the best reference point elevation value that matches the actual point cloud. To reduce initial estimation error, we perform a discrete grid search. We initialize a number of height errors on the reference point. For each value, we generate the point cloud and then calculate its distance to the external LiDAR point cloud. To improve the efficiency of distance calculations, we first build a k–d tree from the external LiDAR point cloud, and then use it to find the nearest neighbor within the LiDAR point cloud for each SAR point. The mean distance of all points is used to assess the match degree. The corresponding height error will be considered as the height error for the reference point.

We define a distance threshold and identify radar pixel points that are farther than this threshold from the external point cloud. These outlier clusters are treated as isolated objects, and we re-run step 2 to re-estimate the parameters for them. Afterward, we repeat step 3 to align them with the external point cloud. If they still deviate beyond the threshold distance, we remove them from the results. For pixel points within the threshold, we retain the 3-D coordinates of the nearest point in the external point cloud for further visualization or analysis.

D. Generation of InSAR 4-D Point Cloud

Before this step, initial estimates for phase-related parameters such as height, velocity, and thermal coefficient of scatterers within each urban object have been obtained. We use these estimates to unwarp the interferograms and then run a regular time-series analysis to remove errors and obtain a refined result [27]. The time-series phase models can be properly selected according to the type of object. The residuals after removing the fit phase are then filtered by a spatial-temporal filtering process to retrieve the nonlinear deformation component. Specifically, we apply a low-pass filter by convolution in time with a Gaussian function. The low-pass-filtered signal at the primary time gives an estimate of primary contribution to atmospheric and orbital errors. By subtracting the low-pass-filtered signal from secondary interferograms, we get an initial guess of secondary contribution to the atmospheric and orbital errors. We then low-pass filter this phase spatially with a 2-D Gaussian function. The output can be considered as the secondary contribution to the atmospheric and orbital errors. We remove all the atmospheric and orbital errors from the residual phase. The remaining phase can be considered a combination of the nonlinear deformation component and noise. A temporal low-pass filter is can further applied to remove the noise signal and extract the nonlinear deformation component.

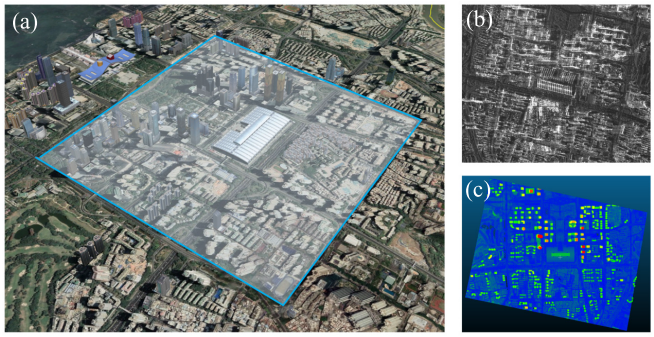


Fig. 3. Study area and data used. (a) Area coverage (depicted by blue rectangle) overlaid on Google Earth, (b) SAR intensity image, and (c) colorized LiDAR data.

III. Results

Our study was conducted in Shenzhen, one of the most populous megacities in China, located in the southern part of the country. This city, with a population exceeding ten million, is densely populated with high-rise buildings and high-density residences. However, its rapid urbanization has made it vulnerable to significant subsidence, leading to a series of infrastructure accidents. For example, on May 18, 2021, a skyscraper mysteriously shook, causing panic and necessitating an evacuation and inspection. Likewise, on December 20, 2015, the improper handling of construction waste triggered a catastrophic landslide, burying 33 buildings and resulting in at least 69 fatalities. On August 28, 2019, a residential building in the Luohu District tilted and collapsed dangerously, damaging staircases in two adjacent buildings, but thankfully, due to efficient evacuation, no casualties were reported.

The study site is situated in the heart of Shenzhen, covering approximately 3.5 km^2 and includes various typical urban features such as high-rise and high-density buildings, overpasses, and more, making it an ideal testing ground for the approach [see Fig. 3(a)]. The dataset for Obj-InSAR consists of 35 COSMO-SkyMed satellite SAR images and a LiDAR point cloud. The SAR images, captured in Stripmap mode with a 3 × 3 m spatial resolution [see Fig. 3(b)], were collected from December 07, 2013, to November 29, 2016. The LiDAR data, obtained using an airborne LiDAR instrument, has a density of approximately 4.09 points/m^2 [see Fig. 3(c)].

A. Urban Objects Segmentation With a Vision Foundation Model

First, we use a foundation model, SAM, to perform semantic segmentation. The input image is a colorized orthographic map

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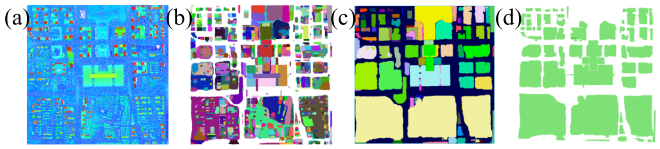


Fig. 4. Multiscale objects segmentation. (a) Colorized orthographic map generated by LiDAR point cloud, (b) objects at all scales as determined by SAM, (c) objects at large scales as determined by SAM, and (d) remaining objects after adjacent objects grouping.

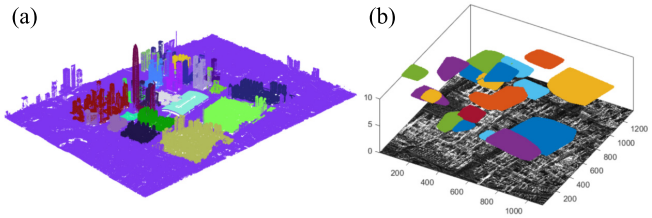


Fig. 5. Segmentation results. (a) Objects in the LiDAR point cloud and (b) objects in the SAR image.

generated by the point cloud using a color ramp shader [see Fig. 4(a)]. We use the point prompt for the segmentation task, whereby SAM uses a point on the object provided by the user as a prompt to segment the object. In this study, we sample 60 points along each side of the image, resulting in a total of 3600 points sampled for mask generation. We set the filtering threshold at 0.7, enabling us to discard any masks that score below this quality threshold. Furthermore, we set a stability threshold of 0.9 to ensure the masks remain consistent under varying conditions. Additional mask predictions are performed on image crops in this setup. We reduce the number of points sampled per side by half for each subsequent layer in our analysis. Lastly, we remove any disconnected regions or holes in the masks smaller than 500 pixels to ensure the integrity of the masks [see Fig. 4(b) and (c)]. In case of overlays between small and large objects, we opt to merge them into the larger one. Ultimately, we obtain 22 objects in the study area [see Fig. 4(d)].

According to the segmentation results, the 3-D point cloud were grouped into 22 categories [see Fig. 5(a)]. The 3-D lookup table can effectively associate spatial points with radar pixels. It was used to link the object segmentation results of the 3-D model in geographic coordinates to the SAR image. It is important to note that due to overlap, some radar pixels may be linked to multiple features (such as building facades, and ground). The Obj-InSAR stores pixels by object, allowing the same pixel to be recorded in multiple object sets, thereby fully preserving information on feature overlap [see Fig. 5(b)].

B. Plug-In Network Construction and Parameter Estimation

Then we shall determine the height and deformation parameters of each object. To eliminate irrelevant signals such as atmospheric noise, a spatial network must be constructed and solved. In our study, we use an adaptive mode named "plug-in" to construct the network. Given our prior knowledge of object segmentation, we can easily define this task by selecting the objects to process according to our needs. First, an object boundary map is plotted in the geographic space, enabling the user to select the objects to process. This selection can include a single object or multiple objects. In the case study, the Exhibition Center building (Object 10) and a residential zone (Object 16) were chosen as examples [see Fig. 6(a)]. Second, we construct a global sparse network as a reference, using only high-quality scatterers [shown in blue in Fig. 6(b)]. Then, local dense networks are built for selected objects. Since the dense network is confined to a very limited area within the object, the computational time and memory usage for further network solving will be acceptable. Finally, the local network is plugged into the global network [see Fig. 6(b)], and they are solved together to get the final results. Significantly, the reduced computational burden of this design allows us to solve all arcs globally without dividing the area into many small patches, thus avoiding the risk of error propagation between patches.

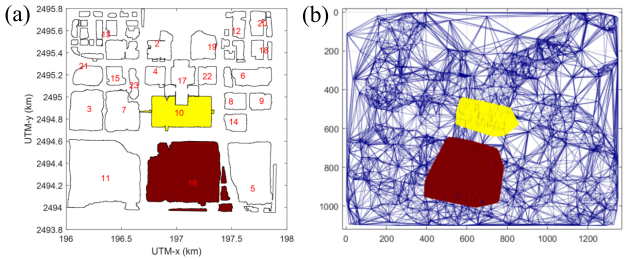


Fig. 6. Plug-in network construction. (a) Object boundaries shown in geographic reference system and (b) dense network constructed for selected objects. Yellow and red colors highlight the two selected objects.

For comparison, we present results from a less dense network. In this case, it is generated in the same manner but using Delaunay in the network construction, compared to our dense network. The Delaunay network comprises 180 251 arcs [see Fig. 7(a)], while the dense network has 4 958 226 arcs [see Fig. 7(c)]. It is shown that the dense network within an object can significantly increase the number of points. The available points increase from 6082 [see Fig. 7(b)] to 25 377 [see Fig. 7(d)] for the residential zone (Object 16). The larger number of points allows more detailed measurement of the objects. However, for traditional processing without defining the objects, we cannot construct a dense network in the same manner because the arc number would be prohibitively high when all pixels are used. Nevertheless, for segmented InSAR, a dense network is possible because it is limited to the object boundary, and detailed InSAR measurements can be obtained accordingly. If all 22 objects in this study area are processed together, there will be 24 869 960 arcs, which might require a substantial amount of computational time and memory. However, the plug-in design of our approach allows for flexible processing of both single and multiple objects, which can generate results as needed.

C. Rectified Height With 3-D Model

The segmentation information establishes a link between the SAR pixels and the 3-D structures of objects in the real world; therefore, enabling the refinement of the initial solution. In our strategy, we first perform ortho-rectification of the scatterers using their estimated heights from SAR observations. The process of converting radar coordinates to geographic spatial coordinates is affected by various error factors, such as satellite orbit, LiDAR height, coordinate reference, and reference point

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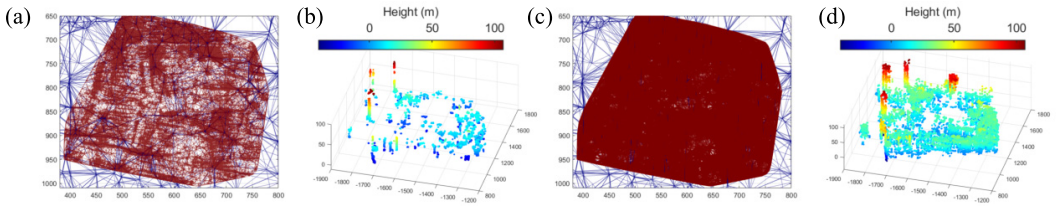


Fig. 7. Height estimation comparison with different network. (a) Ordinary Delaunay network for selected Object 16, (b) scatterers kept using Ordinary Delaunay network, (c) dense network for selected Object 16, and (d) scatterers kept using dense network.

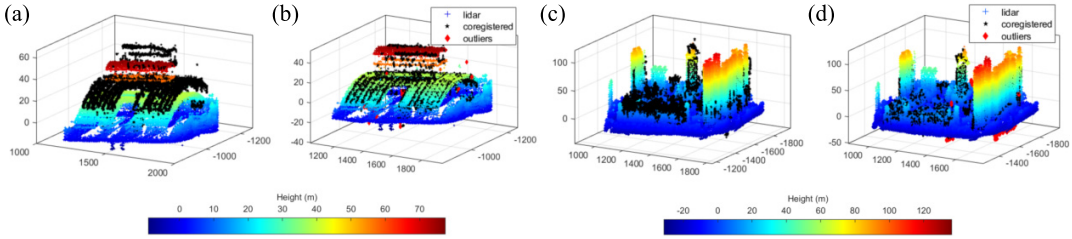


Fig. 8. 3-D point cloud registration and outliers detection. (a) and (b) Point clouds of Object 10 before and after registration and (c) and (d) point clouds of Object 16 before and after registration.

elevation. Among them, the elevation error of reference point is relatively large and affects both the horizontal and vertical positions of the object. Due to the inaccurate initial estimation of the reference point height, the initial SAR point cloud cannot match the external LiDAR point cloud [see Fig. 4(a) and (b)]. To reduce this error, we perform a discrete grid search. After searching for the optimal starting point height, the two point clouds match quite well (see Fig. 8).

Once the constant height error is fixed, we use the distances calculated for each SAR scatterer point to assess the accuracy of their height estimation. As noted in many previous studies, improper network solving or spatial unwrapping could cause significant errors on the height estimation [28]. Many attempts have been made to remove or correct these unnecessary outliers. In our approach, by taking advantage of the external information provided by the segmentation and object-level matching, it is relatively straightforward and accurate to identify and remove these outliers. We used $\pm 3 \times$ standard deviation as the criteria to detect outliers. For each object, we find some points that deviate from the real object geometry. Although each arc was selected based on its temporal coherence, many other factors may lead to miscalculation. These include unreasonable arc connections [28], pixels immediately adjacent to strong PS pixels being mistakenly picked [27], and so on.

We overlaid the outliers in the second object (Object 16) on the multi-look image [see Fig. 9(a)], the retrieved height map [see Fig. 9(b)], and the network [see Fig. 9(c)]. These pixels include independent building scatterers, scatterers located on the tops of neighboring high-rise buildings, and some random scatterers. Although their height values are not correctly estimated, the arcs connecting these outliers still show high coherence. This indicates that high coherence does not guarantee completely accurate estimation. However, a clear characteristic of these outliers is that there are a limited number of arcs connecting them to neighboring point clusters [see Fig. 9(c)]. Their positions are strongly influenced by

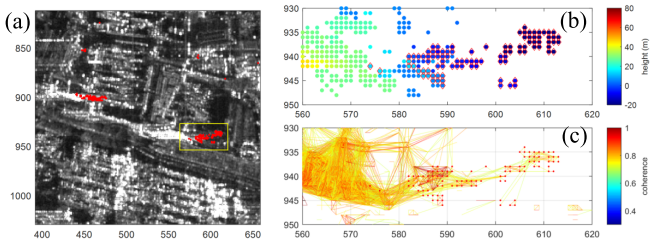


Fig. 9. Outliers overlaid on (a) SAR image, (b) retrieved height map, and (c) network.

the parameter estimation of these arcs. If these key connecting arcs are not accurately solved, the errors will directly propagate to the entire point cluster, as there are no other constraints.

D. Characteristics of Outliers and Reprocessing

High coherence implies good agreement between observation and model. Fig. 9(c) shows that the coherence is not low for the arcs connecting outliers, but the parameters are wrongly estimated. This suggests that the observation cannot strongly constrain the parameters, and the time-series phase observations might be explained by different models. To explore this, we plot the coherence of the two largest coherence solutions during the periodogram search [see Fig. 10(a)]. We find that there are several arcs with high coherence (>0.6) for the second solution. We randomly select one such arc and plot the coherence map for parameter grid searching [see Fig. 10(a)]. There are two significant peaks with coherence higher than 0.6, implying more than one candidate solution. For both solutions, we explore the fit between the simulated phase and the observation [see Fig. 10(c)] and find that both solutions fit the observations very well, making it difficult to distinguish between them.

Several possible reasons may explain this phenomenon. In urban scenarios, there may be many strong scatterers

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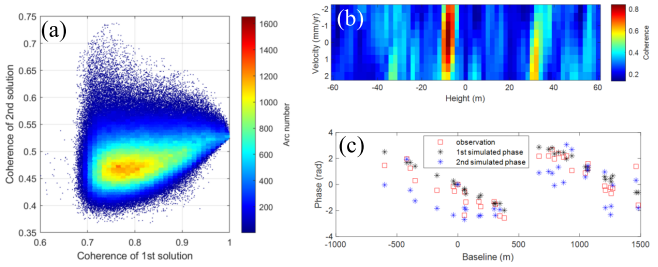


Fig. 10. Illustration of multiple solutions in parameter estimation. (a) Density map of the arc coherence distribution for the two largest coherence solutions, (b) coherence map for parameter grid search of a selected arc with multiple solutions, and (c) fit to observation for two candidate solutions.

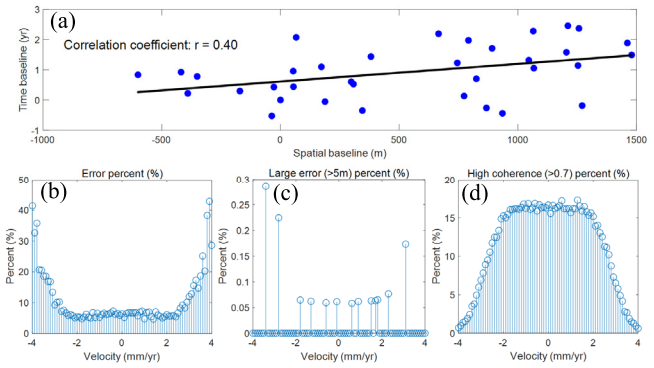


Fig. 11. (a) Correlation between time baseline and spatial baseline, (b) error percentage for high coherence (>0.7) cases in the synthetic test, (c) percentage of large errors (>5 m) for high coherence (>0.7) cases in the synthetic test, and (d) percentage of high coherence (>0.7) cases in the synthetic test.

overlaid within a single pixel. These scatterers have different locations and deformations, causing the observation to be a combination of different mechanisms, which can be partly explained by different parameter sets. Additionally, another possible reason is the correlation between the time baseline and the spatial baseline. We plot these two baselines in Fig. 11(a) and find a nonnegligible correlation between them. To explore the influence of this correlation, we simulate 10 000 time-series phase observations by adding 1 rad of standard deviation noise. The parameter search range for velocity in the periodogram is from −2 to 2 mm/yr, with a 0.5 mm/yr spacing. We run the periodogram with various velocity settings and calculate the percentage for both high coherence estimation and height estimation errors, including significant height estimation errors. It was found that when the true velocity parameter falls outside the search range or is not located exactly on the search grid, the estimation errors significantly increase. In such cases, a significant component of the phase due to velocity is not properly fit. However, this residual phase is mitigated by the height parameter, leading to a significant bias in the parameter estimation. If the time baseline and spatial baseline were completely uncorrelated, such mitigation would not occur.

In a typical InSAR processing chain, there is no reliable way to avoid these errors due to the limitations of the data itself. Fortunately, in our proposed framework, we are able to detect these outliers through co-registration with an external point cloud. This allows us to avoid the influence of abnormal parameter estimation on these arcs. We create

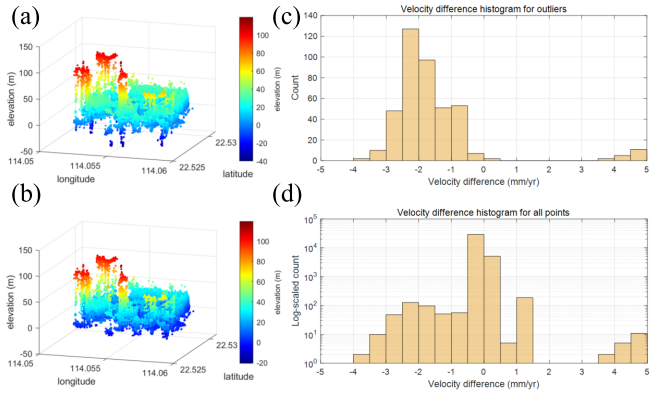


Fig. 12. (a) Original PS point cloud, (b) refined PS point cloud after fine processing of outliers, (c) estimated velocity difference histogram for outliers, and (d) estimated velocity difference histogram for all points.

an isolated object for each outlier point cluster and repeat steps 2 and 3, integrating the new object into our network. Afterward, we retain these pixels and achieve better results. As shown in Fig. 12, the new PS point cloud is clearly more accurate than the original one, which was influenced by defective arcs. Most underground points in Fig. 12(b) have been corrected to the right positions. In addition to the height parameter adjustments, we also observed changes in velocity parameters. These changes mostly occurred in the outliers, with the range of velocity parameter adjustments reaching up to 5 mm/yr for these outliers.

E. 4-D Urban InSAR Measurement

The initial estimates for phase-related parameters can be used to unwarp the interferograms, and a regular time-series analysis can then be conducted to obtain the time-series deformation results. Fig. 5 shows the final Obj-InSAR measurement results, where the scatterers are overlaid on Google Earth (see Fig. 13). For each object, we present a linear velocity, a thermal coefficient, a 3-D position, and a time-series deformation. Since we have performed co-registration for SAR scatterers, the 3-D position is quite precise, and the deformation results can be precisely geolocated to real-world positions. This overcomes geolocation problems caused by geometric errors from satellite orbit, elevation accuracy, and coordinate [29]. It allows us to better visualize and interpret these deformation signals by overlaying the results with other geodata.

Meanwhile, we successfully retrieve the deformation pattern related to long-term displacement and periodic temperature-related deformation. We observe that temperature deformation dominates the deformation pattern of the exhibition building (Object 10). It is recognized as the largest single edifice in Shenzhen, which features a stunning combination of steel and glass at its pinnacle. The glass, possessing a certain thermal expansion coefficient, is highly sensitive to fluctuations in temperature. We further explore the time-series deformation behavior for a selected point located in the Convention and Exhibition Center (see Fig. 14). We observe a clear seasonal pattern that strongly correlates with local temperature changes. On a yearly basis, temperature variations may lead to a deformation range of 10–15 mm. For the long-term behavior,

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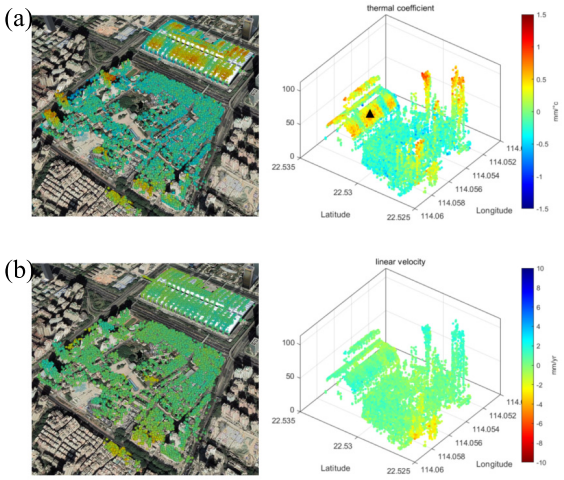


Fig. 13. Deformation products of Obj-InSAR. 3-D visualization of (a) thermal coefficient and (b) linear velocity.

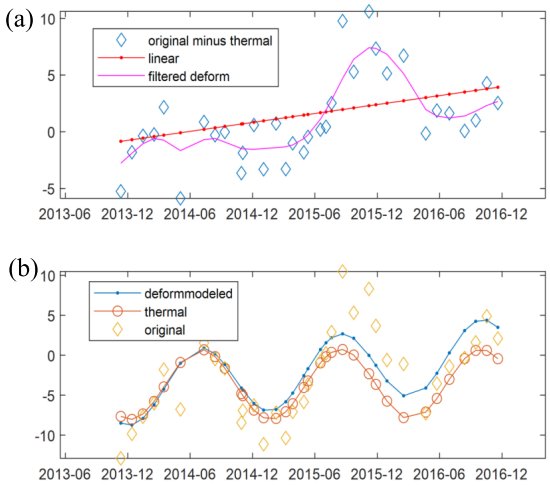


Fig. 14. Temporal pattern of deformation. (a) Time-series deformation results for the selected point [black triangle in Fig. 13(a)] after removing thermal deformation and (b) time-series deformation results for the selected point with thermal deformation.

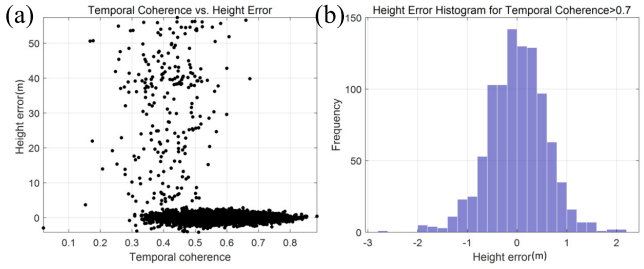


Fig. 15. Height estimation errors using simulated noise. (a) Temporal coherence versus height error and (b) height error histogram for temporal coherence >0.7.

a gradual upward trend, approximately 2 mm per year, might also be observed.

For the residential zone (Object 16), the entire area also experiences minimal deformation, except for some slight subsidence observed on the southern border. The temperature deformation also dominates the deformation pattern, with high thermal deformation occurring at tall locations. The correlation

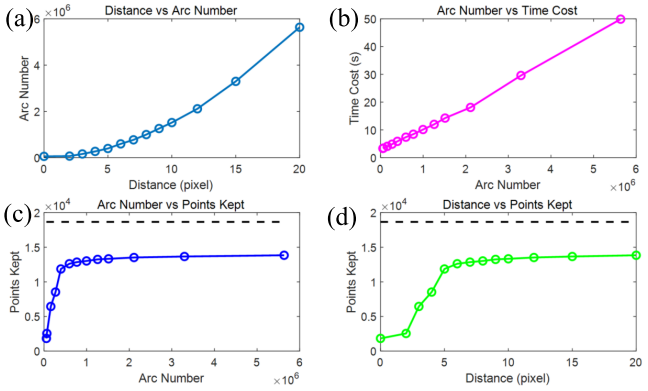


Fig. 16. PSs retained as a function of arc density. (a) Distance versus arc number, (b) arc number versus time cost, (c) arc number versus points kept, and (d) distance versus points kept. Dashed line in (c) and (d) indicate the number of initially selected candidate PSs.

observed is most likely attributable to the cumulative impact of thermal expansion in the vertical direction [30]. Due to the fixed foundation and the unrestricted top, as the temperature rises, expansion of each concrete element at varying heights will accumulate upward, leading to the greatest movement at the pinnacle of the structure in the upward direction. This results in motion toward the sensor in the line-of-sight (LOS) direction. Conversely, when the temperature drops, concrete elements contract, causing the greatest movement to occur at the building's top, but in the downward direction. This leads to movements away from the sensor. Consequently, the deformation range tied to temperature variations is highest at the apex of the structure and diminishes to zero at the base.

IV. Discussion

The rapid increase in the number of SAR satellites, together with improvements in image resolution, highlights the need for an InSAR processing framework that is adaptable, accurate, and efficient, especially in urban environments. InSAR technology has proven successful in various fields, but urban monitoring presents unique challenges due to the complexity of these environments and specific engineering standards. Urban environments are often high-density and display significant spatial variation, resulting in more complex pixel components with various types of scatterers. These complexities disrupt standard assumptions that are based on homogeneity. Additionally, engineering standards for deformation monitoring in these high-density urban areas require greater point densities, more accurate positioning, and precise deformation measurements. The strategies for geocoding scatterers and parameter estimation in conventional InSAR processing may not be adequate. Therefore, monitoring surface deformation in high-density urban environments poses a novel challenge that necessitates advancements in InSAR theoretical technology. Our proposed Obj-InSAR method facilitates precise object-oriented analysis, offering a significant improvement over conventional pixel-oriented approaches.

A. Integrating Additional Constraints From Other Geodata

The first benefit of our proposed Obj-InSAR is the addition of constraints from external geodata. The inherent conditions

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of SAR observation already set the upper limit of result quality. According to the theory of statistical estimation, there is always a lower bound on the variance of unbiased estimators, given the function model and stochastic model. Existing SAR data observation cannot constrain a very precise height estimation. A large height estimation error can affect the positioning of scatterers, which in turn impacts the accuracy of geocoding. It is illustrated in Fig. 4 that the height of many scatterers was wrongly estimated. Using this incorrectly estimated height, these scatterers were positioned underground, leading to unreasonable results. Fusion of multisource data is a routine method in geodata processing; however, it is rarely seen in urban SAR processing due to the complex imaging geometry of SAR.

Obj-InSAR first utilizes a 3-D lookup table to link external data to SAR data. It then implements a co-registration step based on the semantic information of objects to rectify the location of scatterer clusters. This allows us to identify and remove outliers that may contaminate overall InSAR quality. In urban areas, the complex structure may cause sudden changes in height between neighboring scatterers [28]. Solving height parameters on these arcs is quite challenging. We simulated a set of noise-affected data using the observational conditions of the experimental area (see Fig. 15). It was found that, even under the selection with a temporal coherence threshold of 0.7, there is still a significant number of arc height estimation errors reaching 2–3 m (see Fig. 12). Errors from poor estimation of arcs across long distances can easily be propagated to the point in the network adjustment. Our proposed Obj-InSAR uses external data to increase the information available for processing; thus, providing a more robust solution than one relying solely on SAR data. As illustrated in Figs. 10 and 12, outliers are identified by registering with an external point cloud. By disconnecting the defective arcs and constructing new networks, a more accurate solution is achieved.

B. Object-Based Flexible Area Selection and Computation

Unlike measuring natural surface deformation, urban deformation measurement involves different structures and surfaces such as buildings, bridges, roads, and bare soil. On the one hand, different surface objects in urban scenarios exhibit distinct deformation patterns [31]. For instance, steel structures may experience significant thermal deformation, and urban underground projects may undergo nonlinear sudden deformations. Therefore, a uniform deformation calculation method based on a single type of deformation pattern will inevitably lead to inaccurate results. On the other hand, for engineering managers, the concern might only be about certain specific structures or the need for periodic, precise deformation results of specific projects. Calculating deformation across an entire city or region can be time-consuming and labor-intensive, and often unnecessary. As deformation monitoring shifts from large-scale natural surfaces to detailed urban scenarios, a flexible and configurable InSAR measurement method is required.

Obj-InSAR offers such an object-based processing strategy, constructing a hybrid network consisting of a sparse global network and a dense object network. The object network can be plugged in and out of the global network as needed. This plug-in network design allows for special and flexible handling of interested structures or construction sites. It combines the advantages of detailed processing in small areas with the global scope of large-scale processing. Each object can choose an analytical method that suits its characteristics, and the results for each object can also be connected to the larger-scale sparse network to ensure the integrity of the solution. Of course, this method can also significantly save on computational time while ensuring measurement quality.

C. Efficient Network Construction for Detailed PS Measurements

In urban scenarios, achieving more detailed measurements requires denser InSAR data. Effective techniques can significantly increase the point density in InSAR measurements. For example, TomoSAR efficiently utilizes overlapping scatterers, while phase linking optimizes the phase of distributed scatterers. Beyond adding back overlapping and distributed scatterers with advanced methods, many PSs that are often inappropriately excluded from analyses can also be reintegrated. By restoring these unnecessarily discarded points, the observational density of the data can be substantially enhanced.

In PS-InSAR, initial PS points are often determined based on the amplitude dispersion criterion, followed by the use of temporal coherence to finalize the suitable PS points. Temporal coherence is notably affected by atmospheric conditions; if an arc has a significant atmospheric component, its temporal coherence will be low, leading to poor quality in arc parameter calculations and its unsuitability for network adjustment. If no arcs with high temporal coherence exist on a PS point, that point will be discarded. To increase the retention rate of PS points, it is essential to establish as many connecting arcs as possible for each PS point. However, a fully connected network in previous methods is a big challenge because it would generate a large number of arc segments which are difficult to handle. Many studies have indicated that the cost for dense network is significant in terms of computer memory requirement and processing time [32]. Typical networking methods may not calculate all possible connections, meaning not all PS points are preserved to their fullest extent. Several attempts to solve this have been proposed [33]. One widely used approach is based on decomposition in spatially adjacent blocks [16], [34]. While it works in many scenarios, this approach can be affected by unstable connections and constraints, and local errors accumulated in patch-to-patch propagation can easily and visibly cause patch effects. Another method is based on decomposing the problem at different scales according to a pyramidal scheme [35]. However, this requires additional processing of multilooking data, and in an urban scenario, the lower scale data may not be precise enough to be taken as control points to constrain higher scale solutions.

Our method effectively manages this workload. It treats each object as a processing unit. Compared to the entire area, an object comprises relatively fewer points, allowing for the construction of fully connected arcs. Additionally, objects segmented under geographical coordinates inherently have spatial continuity, and the internal connection space of an object is likely to have lower atmospheric variability, ensuring some level of temporal coherence in arcs. By connecting all possible point combinations, this approach can retain the maximum number of points possible, thereby achieving high-density InSAR measurement results. We examine the Object 10 shown

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in our case study (see Fig. 6) as an example to demonstrate the impact of arc density on the number of points kept. In the test, we start with the Delaunay network as the default and progressively add fully connected arcs within different distance ranges to the network. It is evident that these incrementally added arcs significantly increase the number of PSs that can be retained (see Fig. 13). Naturally, as the distance threshold increases, so does the number of arcs explosively. However, in this case, it can be observed that expanding the distance threshold to seven pixels already retains a high number of PSs without necessitating a substantial computational effort (see Fig. 16). Additionally, with parallel computing and the design of sparse matrices, approximately 10 000 arcs can be solved per second, demonstrating that the Obj-InSAR approach, which is based on objectwise fully connected networking, is feasible for actual urban InSAR analysis.

D. Future Extensions and Prospects

Besides LiDAR point clouds, it is also possible to include optical point clouds in our framework. Unmanned aerial vehicles (UAVs) make the generation of optical point clouds more flexible and affordable. Although the precision of such point clouds is not as high as LiDAR, they typically offer better spatial resolution and include RGB information. In our current framework, we primarily rely on external 3-D structures to detect defective arcs and guide the unwrapping path, so very high precision point clouds are not essential. However, we acknowledge that this study is pioneering work in processing InSAR on an object unit. Several avenues for improvement could enhance the performance of Obj-InSAR. For example, our work is currently limited to integrating 3-D point cloud data, but other geodata, such as optical or multispectral imagery, could provide additional semantic information. This data could assist with object segmentation, guide network construction, constrain parameter estimation, help select the appropriate phase model, or aid in interpreting deformation results.

V. Conclusion

In this study, we introduce a novel Obj-InSAR processing framework. This framework is based on object units. Our Obj-InSAR framework presents a practical solution for implementing object-oriented SAR interferometry. We leverage the pre-trained foundation model, SAM, to provide initial 3-D structural segmentation data, eliminating the necessity for extensive training while ensuring favorable results. This segmentation information is then used to design an adaptive SAR interferometry-processing framework. Our unique approach entails a plug-in network layer construction that perceives objects as modular entities; hence, allowing for their effortless insertion or removal. The dynamic nature of the network construction algorithm empowers users with the flexibility to modify the object network density based on their requirements. Furthermore, the well-established InSAR time-series analysis procedure can be readily adapted to fit within this framework; thus, guaranteeing measurement results of high precision and reliability.

Our case study demonstrates that the Obj-InSAR approach provides a more accurate, faster, and adaptable solution for urban SAR interferometry, effectively addressing the challenges of SAR processing in urban settings. Outliers resulting from incorrect estimations were successfully identified and corrected. The density of PS points was significantly improved compared to conventional methods, and the computational efficiency is acceptable for practical use. Looking forward, we expect that for very high-resolution SAR data, Obj-InSAR will enable detailed analysis at a fine scale. This could be particularly beneficial for specific man-made structures like bridges and overpasses. The ability of Obj-InSAR to explore intricate details could allow for the integration of InSAR measurements with mechanical numerical modeling, enhancing safety analysis beyond deformation measurement.

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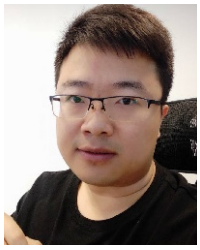
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